Highlights

- Dual-port, $0.054\lambda_0$-thick SIW slot array for the 28 GHz 5G band on a single RT/duroid 5880 layer.
- Resonant slits in the shared top wall, dimensioned by characteristic-mode analysis, raise the port isolation.
- Full-wave slot characterisation shows an order-of-magnitude larger conductance than the Stegen formula in thin SIW.
- Open-source Numba FDTD + RWG MoM/characteristic-mode environment; no commercial EM software needed.
- All solvers, scripts and data are released for full reproducibility.